

- 2) 12 samples of data (six each for the right and left) are taken as a frame for signal processing. In other words, 24-symbol data constitutes a frame. W1 to W24 shown in Fig. 4-3 correspond to it.
- 3) Data W1 to W24 are first scrambled by C2. Scramble may be considered to be a small interleave. In C2 scramble, the data are delayed over two frames alternately in terms of sample data.
- 4) After C2 scramble, the data enter the C2 encoder, which generates 4-symbol parity signals Q1 to Q4 from the 24-symbol data using reed solomon code, and arranges them in the frame center.

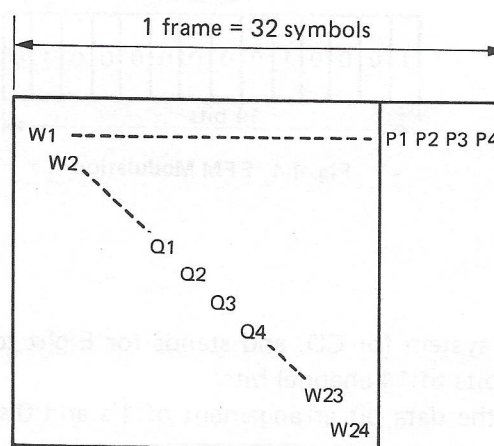
The symbol data count in one frame is 28 in total.

- 5) Next is the interleave.

As shown in the figure, the interleave delays the 28-symbol data, including the Q parity signals, over the range of 4 frames minimum to 108 frames maximum.

- 6) The C1 encoder generates 4-symbol parity signals P1 to P4 from the 28-symbol data, which have been rearranged by interleaving, using reed solomon code, and arranges them at the frame end.

Here, the symbol data count in a frame is 32 in total. (See Fig. 4-3.)



* Symbol parties P1 to P4 are added to the 28 symbols arranged horizontal in a frame, and four symbol parities are added to the 24 symbols arranged diagonally.

Fig. 4-3 CIRC Code Composition

- 7) The C1 scramble delays only the even-numbered symbols of the 32-symbol data in a frame to the next frame.
- 8) Control data P to W are otherwise called subcodes or users bits, containing time information for random access or display and consisting of a single symbol (8 bits).
These signals are arranged at the head of a frame.
Entry of these control signals, increases the symbol data count from 1 frame to 33 in total.
- 9) Finally, the 1-frame data are subjected to EFM-modulated and are recorded on the disk with a frame synchronizing signal is added.
EFM modulation is described in detail in the next paragraph.

All these are intended for improving the correction ability.
This system is called CIRC (Cross Interleave Reed Solomon Code).

(2) EFM modulation

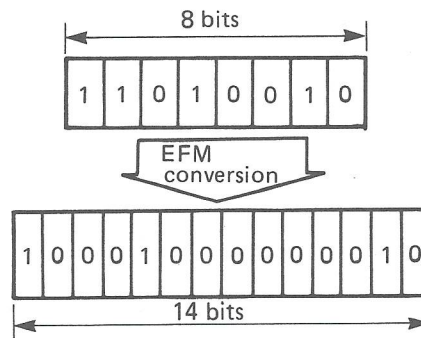


Fig. 4-4 EFM Modulation

EFM is a new modulation system for CD, and stands for Eight to Fourteenth Modulation meaning modulating 8 symbol data bits to 14 channel bits.

EFM is necessary because the data bit arrangement of 1's and 0's cannot be defined if symbol data remains as such.

If 1's or 0's continue so long, data will be a direct current so that self-clocks cannot be generated by the tracking servo (using pits) and reproducing system.

Conversely, if 1's and 0's are frequently reversed, it will be necessary to reduce the laser beam to a very small diameter.

Therefore, signals to be recorded must be such that 1's and 0's are reversed at appropriate intervals and that the DC component is as small as possible.

8 bits has 256 combinations (2^8), while 14 bits has 16,384 combinations (2^{14}).

EFM selects bit patterns, in which 2 to 10 0's are present between two 1's, from among 2^{14} combinations for correspondence to 2^8 data patterns.

Therefore, signals to be recorded have definitions with 3T to 11T (T meaning 1, 1 clock after EFM = 4.32 MHz).

Fig. 9-5 shows the process of up to the recording signal format.

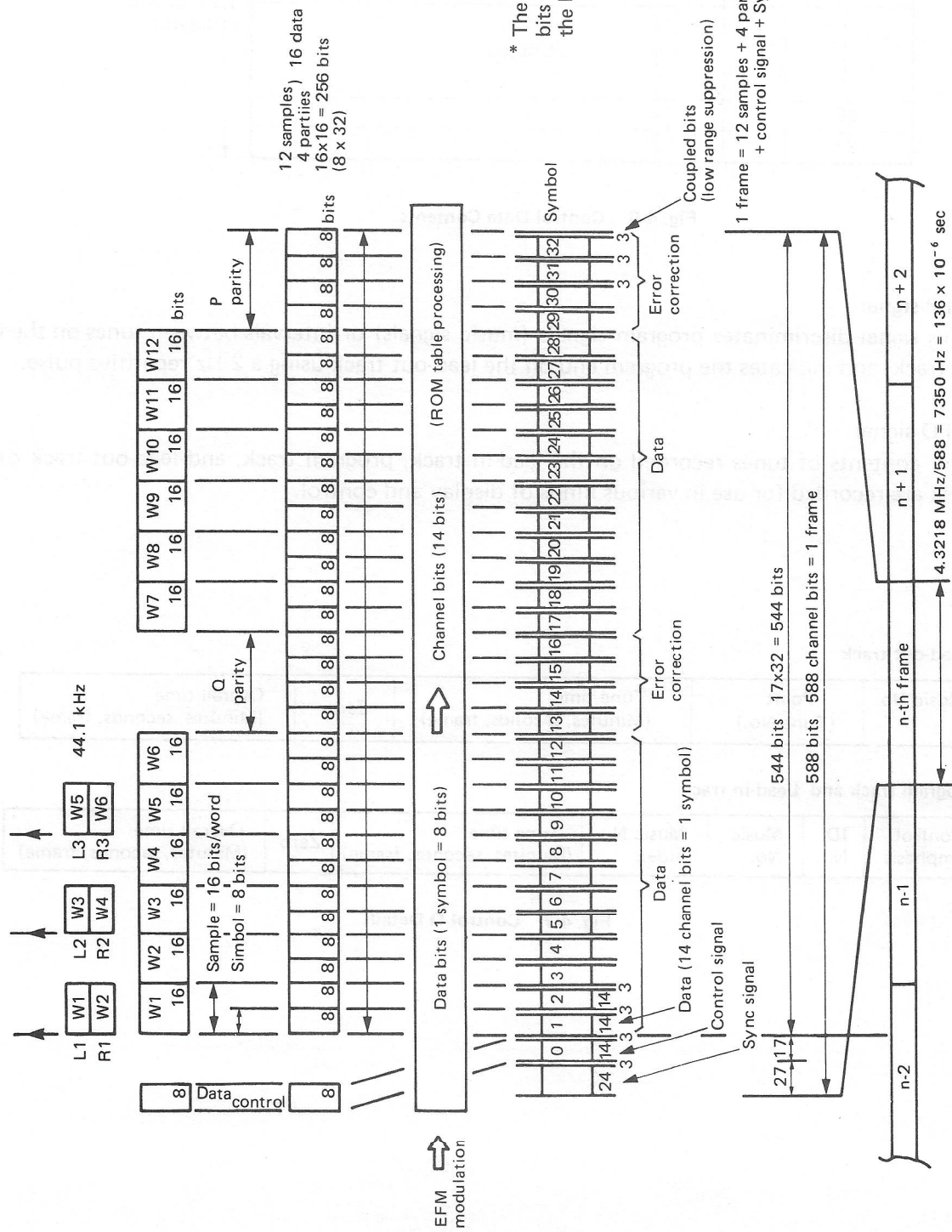


Fig. 4-5 CD Signal Format

(3) Control data

Control data is recorded in one symbol (8 bits) in one frame.

The 8 bits are named P, Q, R, S, T, U, V, and W in this order. Actually, however, only bits P and Q of the 8 bits are used.

Subcode channel No.	P	Q	R	S	T	U	V	W
ϕ	Sync signal							
1	Sync signal							
2								
...								
96								
97								

1/75 second
(13.3 ms)

Fig. 4-6 Control Data Contents

Control P signal

This signal discriminates program signals (music signals) or intervals between tunes on the lead-in track, and indicates the program end on the lead-out track using a 2 Hz repetitive pulse.

Control Q signal

The contents of tunes recorded on the lead-in track, program track, and lead-out track of the disk are recorded for use in various kinds of display and control.

Lead-out track

Music No.	Point (Tune No.)	Tune time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
-----------	---------------------	--	------	---

Program track and Lead-in track

Control emphasis	ID No.	Music No.	Music No. index	Tune time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
---------------------	-----------	--------------	--------------------	--	------	---

Fig. 4-7 Control Q Details

C4-4-1 Reproducing System Signal Processing

(1) Decoder

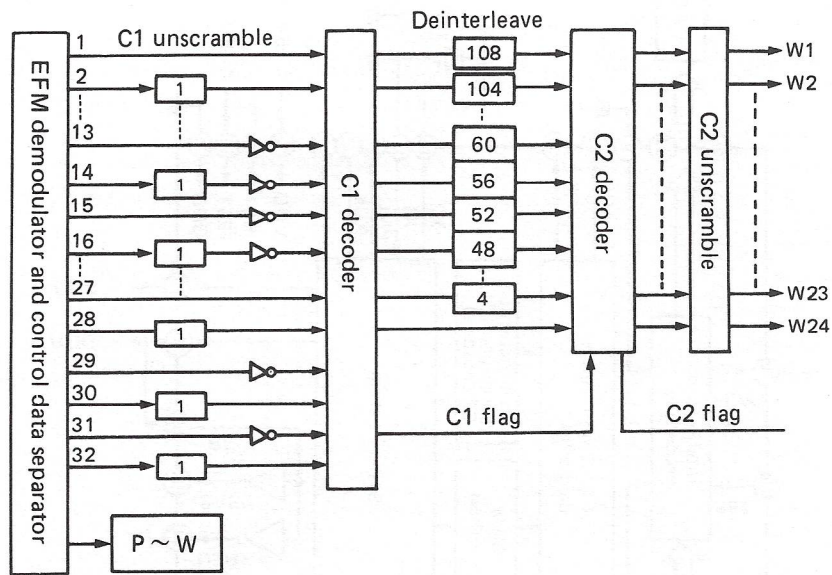


Fig. 4-8 Playback Signal Processing

The flow of playback signal processing is just opposite to the flow of recording system signal processing. Here, decoders for error detection and correction are used in place of the C1 and C2 encoders that are used for parity generation.

Both the C1 and C2 decoders are capable of error detection and correction. If data that cannot be corrected by C1 decoder is marked with C1 flag, and after deinterleaving, the data is sent to the C2 decoder for accurate correction.

If data that cannot be corrected by C2 decoder, the average value obtained from the adjacent data is switched to interpolation after the process of C2 unscramble.

(2) LSI signal processing

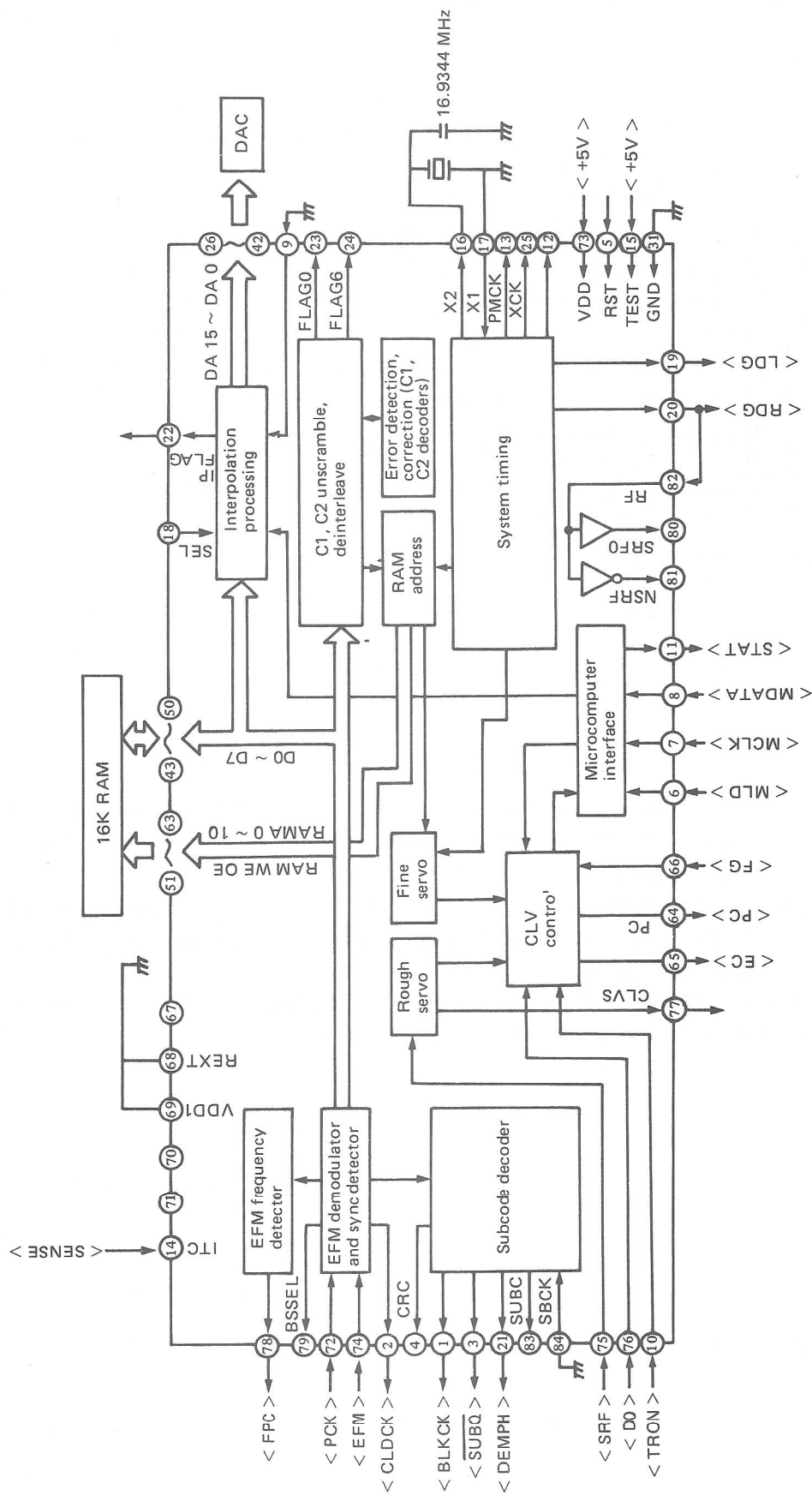


Fig. 4-4 LSI Block Diagram

C4-4-3 LSI and Associated Operations of Peripheral Circuits

(1) Operations associated with AN8371S

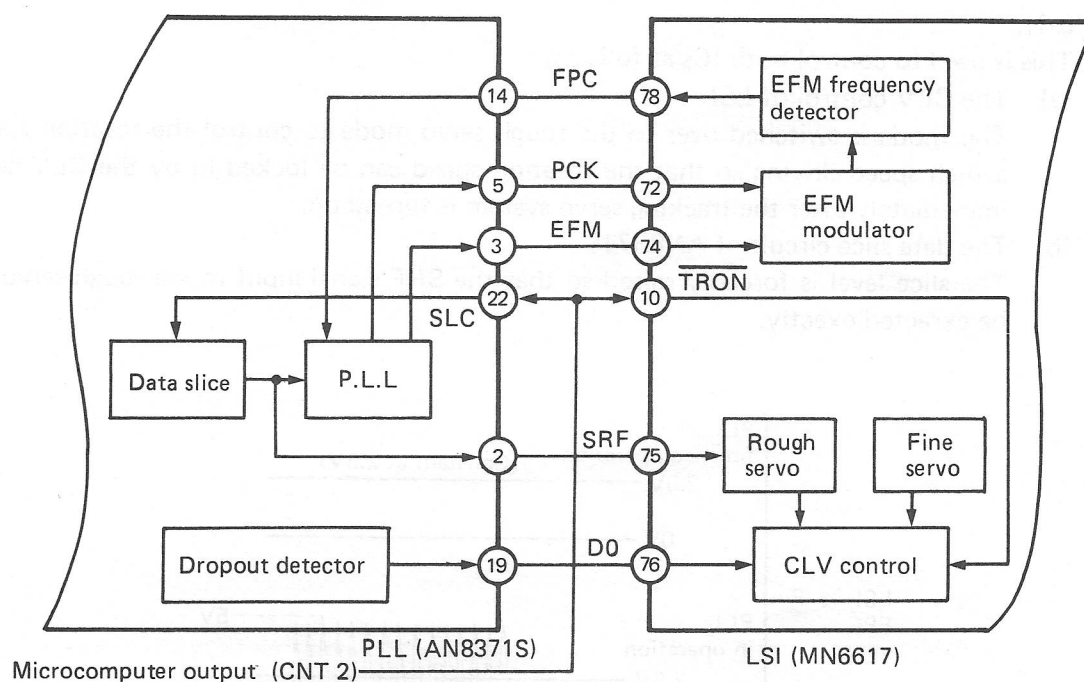


Fig. 4-10 AN8371S and LSI Connection Diagram

Fig. 4-10 shows an LSI (MN6617) and PLL IC (AN8371S) connection diagram.

There are four inputs and one output as viewed from the LSI, and the output (CNT2) from the microcomputer is input to both ICs.

1) PCK, EFM, FPC signals

PCK and EFM signals have been described in Section C4-3.

FPC signal results from measuring the pulse width of EFM signal (3T to 11T) with PCK signal (1T).

If EFM signal is a signal of 2T or less, a feedback operation takes place to raise the VCO frequency of the PLL. If EFM signal is 12T or over, the VCO frequency is lowered.

During play (PLL has completed operating), pin 78 of LSI is constant at 2.5V. If the PLL is being pulled in, a control pulse of $2.5V \pm 2.5V$ is output from pin 28.

2) SRF, D0 signals

SRF signal is generated by data-slicing RF signal (data slice output of AN8371S), and is input to the CLV rough servo (11T servo) of LSI. The CLV servo is described later.

The D0 signal is a dropout detection signal (dropout detection output of AN8371S) of AN8371S, and is used as CLV control signal of LSI. If dropout is detected, the 11T servo is turned off to allow free rotation. Fig. 4-12 shows SRF signal and D0 signal.

- 3) The CNT2 (SLC or TR ON) signal is one of the 4-bit control logic signals of the optical servo IC (AN8370S). This signal remains high until the tracking servo is in the OFF mode (See Table 6-1).

This is used to control both ICs as follows.

- (a) The CLV control of LSI

The mode is switched over to the rough servo mode to control the rotation speed even in a high speed driving so that the rotation speed can be locked in by the CLV servo system immediately after the tracking servo system is turned on.

- (b) The data slice circuit of AN8371S

The slice level is forcibly raised so that the SRF signal input to the rough servo system can be extracted exactly.

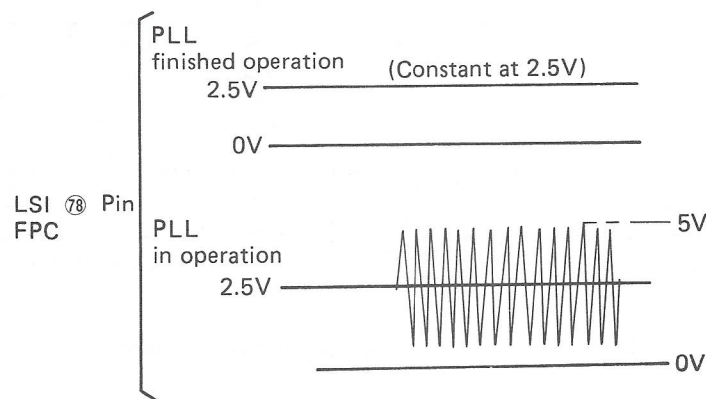
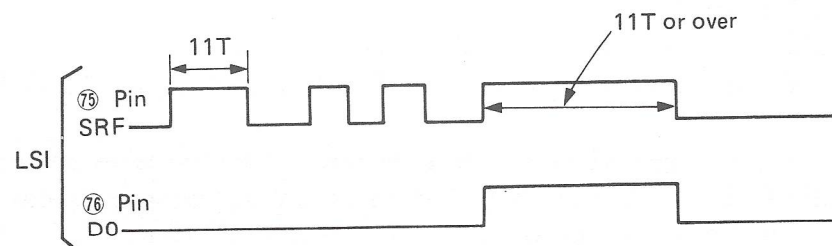


Fig. 4-11 FPC Signal



- SRF signal has a period of 11T or longer when a dropout occurs, which is interpreted as slow rotation, causing the spindle motor to run away. Thus, D0 signal makes rotation free.

Fig. 4-12 SRF Signal and D0 Signal

(2) Operations associated with microcomputer

Fig. 4-13 shows an LSI (MN6617) and microcomputer (MN1554CAF) connection diagram. There are four inputs and five outputs as viewed from the LSI.

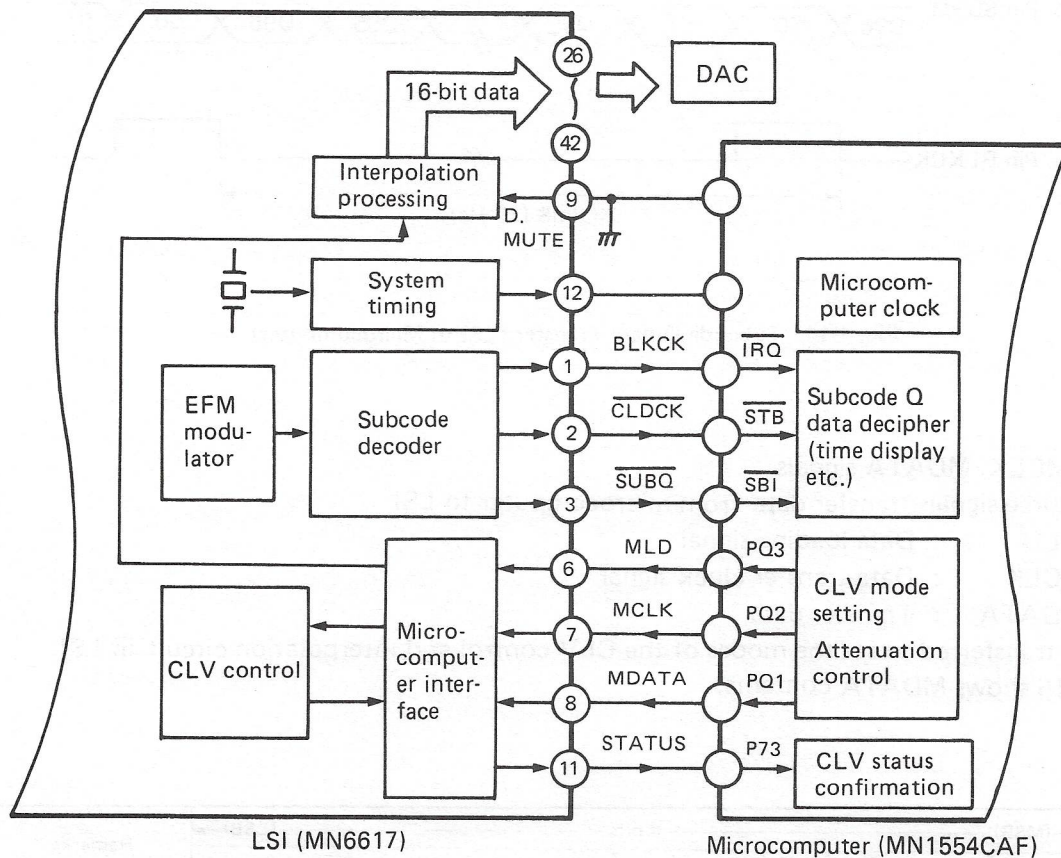


Fig. 4-13 LSI and Microcomputer Connection Diagram

1) BLKCK, CLDCK, SUB Q signal

These three signals transfer the subcode Q data demodulated from the EFM data by LSI to the microcomputer. The microcomputer decodes the subcode Q data and carries out time base control and display.

The BLKCK is a clock signal of one block (98 frames) extracted from EFM data; the CLDCK is a clock signal of one frame similarly extracted, and the SUB Q becomes subcode Q data. Data are transferred serially.

Fig. 4-14 shows a timing chart of the three signals.

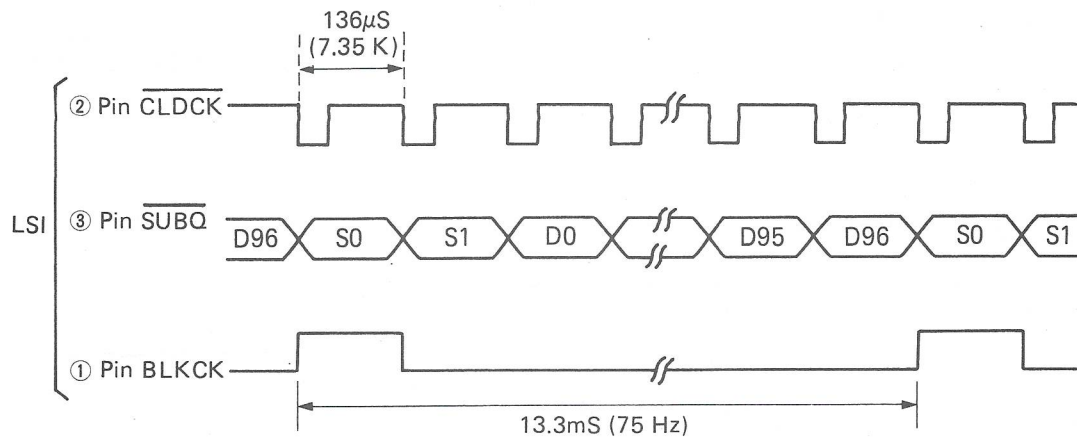


Fig. 4-14 Subcode Q data Transfer (LSI to Microcomputer)

2) MLD, MCLK, MDATA signals

These three signals transfer data from microcomputer to LSI.

- (a) MLD : Data loading signal
- (b) MCLK : Data transfer clock signal
- (c) MDATA : Transfer data

Data is transferred to set the modes of the CLV control and interpolation circuit in LSI.

Fig. 4-15 shows MDATA contents.

No.	8 bits								Remarks
	(MSB)				(LSB)				
	A4	A3	A2	A1	D4	D3	D2	D1	
①	0	0	0	1	1 (with FG)	1 (TT ON)	0	0	CLV mode
							(STOP)		
					0 (without FG)	0 (TT OFF)	0	1	
							(ACC)		
0	0	1	0	—	—	1	0		
						(BRAKE)			
						1	1	(PLAY)	
②	0	0	1	0	—	—	1 (ATT-12dB)	1 (MUTE ON)	Interpolation circuit
							0 (ATT OFF)	0 (MUTE OFF)	
③	0	1	1	1	—	—	0	0	Status output switching
							(CRC)		
							1	0	
								(CLVOK)	
								1	1
								(TT STOP)	

Fig. 4-15 MDATA Contents

The MDATA signal comprises 8 bits, and is serially transferred.

Of the 8 bits, the upper 4 bits express the type of data being transfer data and the lower 4 bits carry data.

Fig. 4-16 shows a timing chart of the three signals.

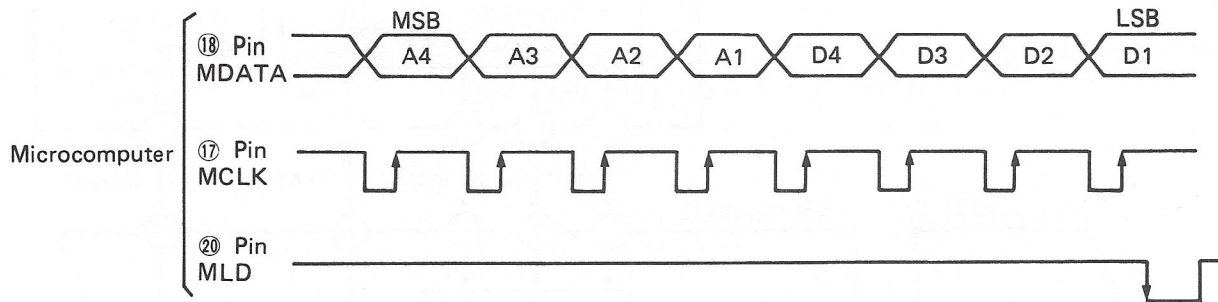


Fig. 4-16 MDATA Transfer (Microcomputer to LSI)

STATUS signal is data to be transferred from LSI to microcomputer, and its contents are checked by microcomputer by switching over the output signal as in Item 3), Fig. 4-15.

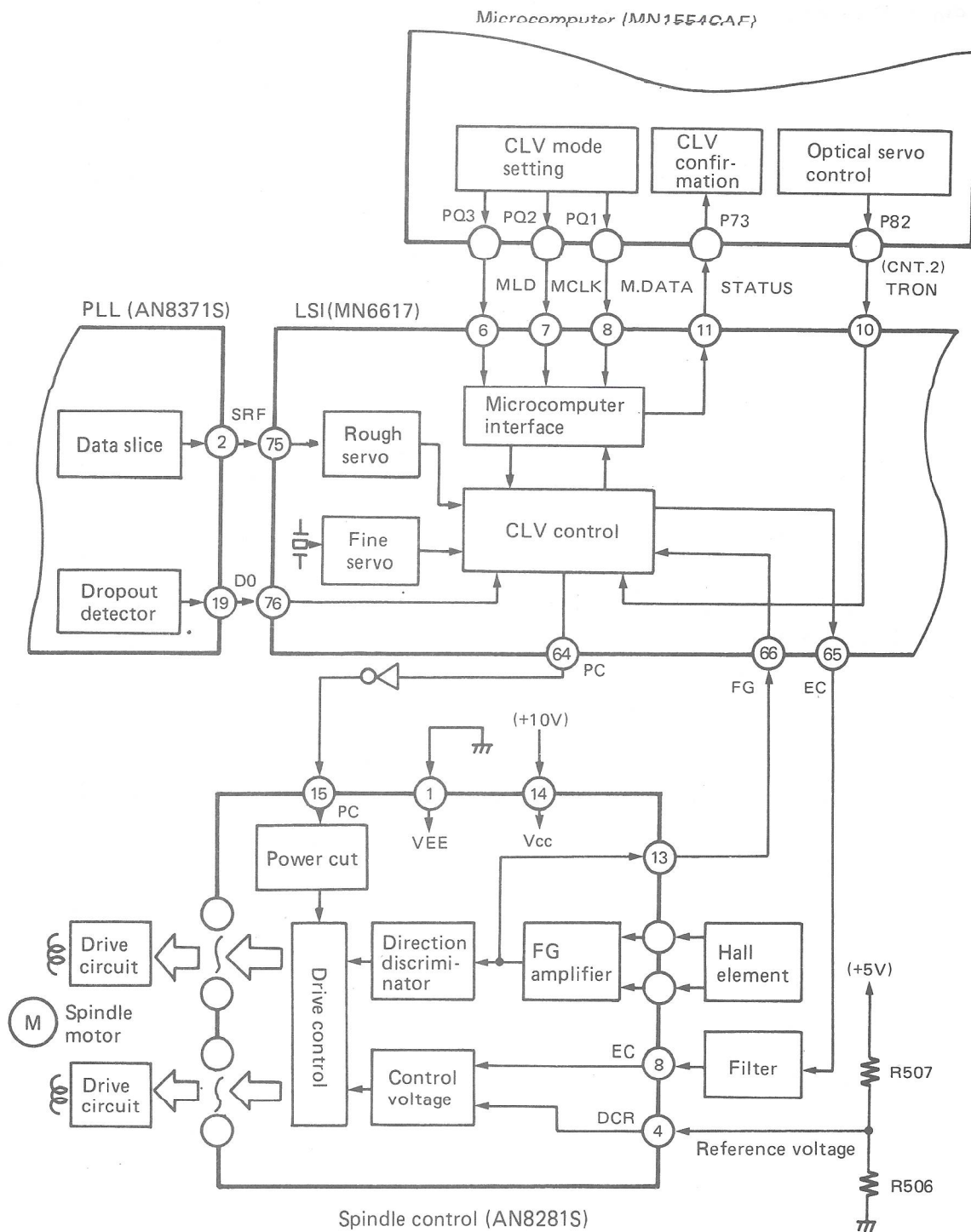


Fig. 4-17 CLV Servo Circuit

The CLV servo is a rotation speed control servo for maintaining linear speed of the disk constant. The servo is divided into three parts, that is 11T servo, CLV servo, and traverse tracking servo. The traverse tracking servo and 11T servo are rough servos, and the CLV servo is a fine servo. The CLV control which is the primary part of the circuit in the LSI. The CLV servo circuits consists of micro-computers AN8371S and AN8281S.

Figure 9-17 shows the CLV servo circuit.

1) AN8281S circuit operation

AN8281S is the spindle motor drive IC. The control voltage EC is applied from the CLV control circuit is applied to pin 8, and the reference voltage divided by R506 and R507 (DCR) is applied to pin 4.

FG and rotating direction are detected by the Hall element, and the FG signal is fed back to the CLV control circuit from pin 13.

The ON/OFF function of this IC (pin 15) controls start and stop of the motor.

Fig. 4-18 shows EC signal versus the motor.

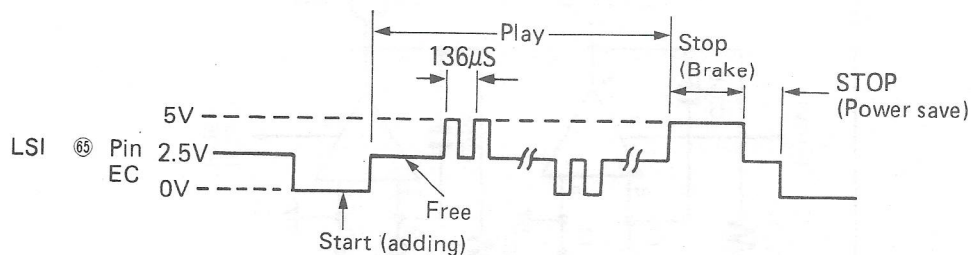


Fig. 4-18 CLV Control Output

2) 11T servo

The 11t servo changes the constant speed to constant linear speed.

This requires some signal extracted from the disk, that is, SRF signal. (SRF signal is generated by data-slicing RF signal.)

The 11T servo controls the SRF signals that the longest period of the signal will be equivalent 11 times the period of the reference clock 1T (4.3 MHz).

When the 11T servo operates, the spindle motor comes close to the control speed of the fine servo.

3) CLV servo

The CLV servo is started by the 11T servo when digital signals are ready for processing in the LSI.

For digital signal processing, data is once written in an external RAM and read out again. A RAM address generator circuit is provided in the LSI for this purpose. Because a crystal clock is used for reading data and a clock synchronized with reproducing signals (corresponding to PCK) is used for writing data, the address locations fluctuates in RAM. The fine servo detects the address location signals and operates to keep the read/wirte pace constant.

4) Traverse tracking servo

The traverse tracking servo follows the pick-up and controls the linear speed of the pick-up constant when the traverse mechanism at high speed.

This servo operates in such a way that two continuous minimum periods of the SRF will be 6 times the period of the reference clock 1T (4.3 MHz).

TRON is used for switching the traverse tracking servo to the 11T servo or vice versa. That is, the traverse tracking servo is selected when TRON is high and the 11T servo is selected when TRON is low.

is low.

AUDIO CIRCUIT

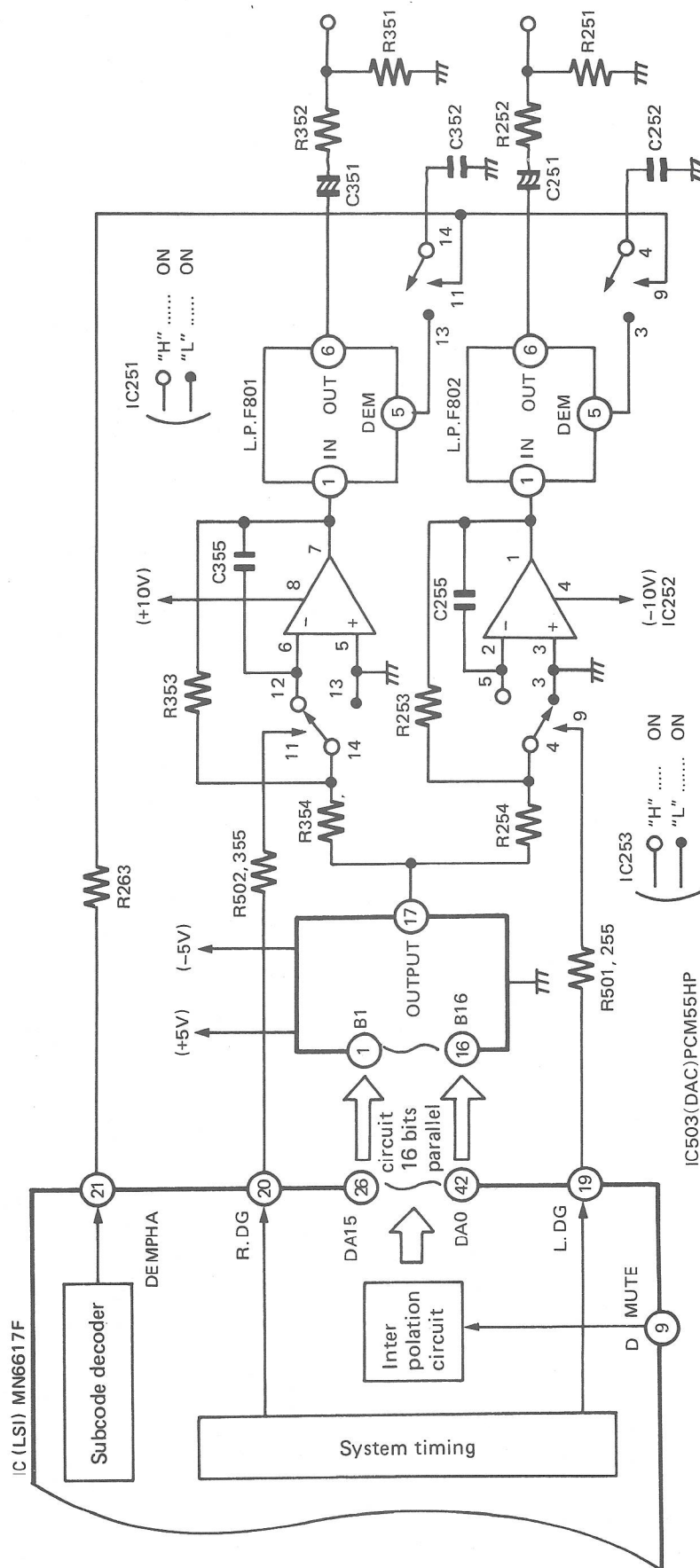
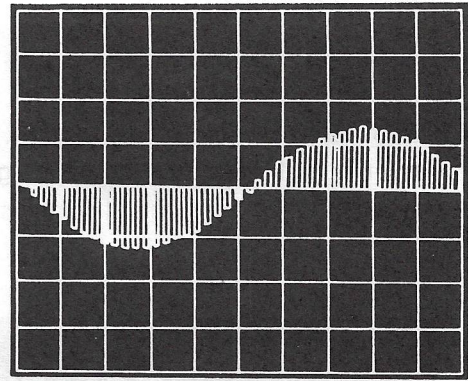


Fig. 4-1 Audio Circuit

C4-5-1 D—A Converter

IC503, is a D—A converter, which converts digital signals to analog signals. Digital signals are input to pins 1 to 16 of IC503, 16 bits parallel. Pin 1 (B1) is MSB, and pin 16 (B16) is LSB. PAM signal is output from pin 17 (OUT).



(Lch = 1 kHz, Rch = 0)

Fig. 4-2 PAM Signal

C4-5-2 Deglitch Circuit

The deglitch circuit consists of IC253 (analog switch), IC252 (operational amplifier), C255 and C355 (for hold).

Pin 19 (L DG) of IC601 (LSI) is the deglitch signal output for the left channel, and pin 20 (R DG) of IC601 is the deglitch signal output for the right channel.

Fig. 4-3 shows a demodulation timing chart of the deglitch circuit.

When L DG signal is low, the left channel circuit takes signal in, while the right channel circuit outputs the signal that have been taken in and held by C225.

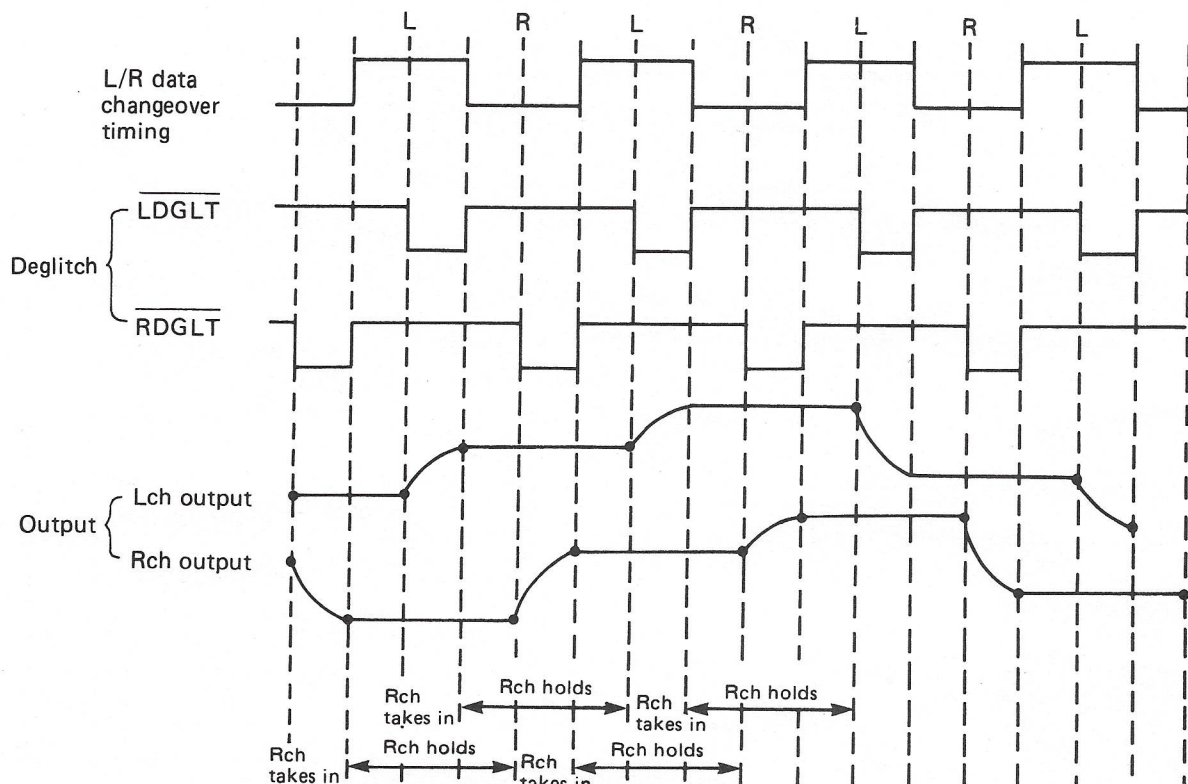


Fig. 4-3 Delitch Timing

C4-5-3 LPF and De-emphasis Circuit

The lowpass filter (LPF) restores the signals containing harmonics (step waves) received from the deglitch circuit to a complete audio signal.

The LPF also has a de-emphasis function. It reads emphasis or no emphasis from subcode (Q channel) recorded on the disk, and the control signal is output from pin 21 (DEMPHA) of IC601 (LSI). De-emphasis is applied when the control signal is high.

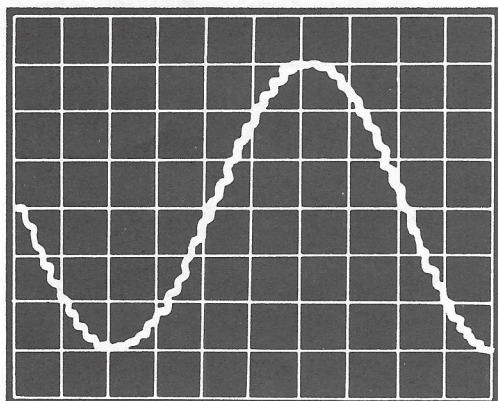


Fig. 4-4 LPF Input

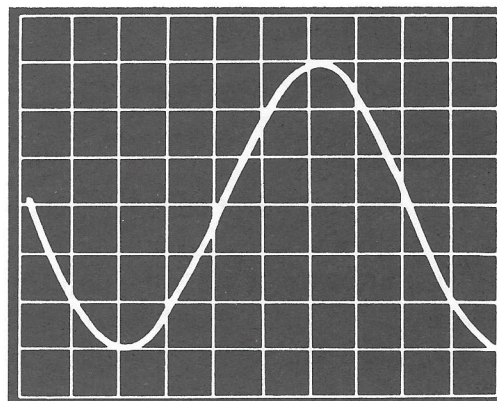


Fig. 4-5 LPF Output

ALIGNMENT INSTRUCTIONS

A5-1 ELECTRICAL ADJUSTMENT

A5-1-1 Measuring Instructions and Tools

1. CD gain alignment jig YESFZ5019
2. Two-channel oscilloscope (30 MHz or over)
3. Low-frequency oscillator
4. Screwdriver for adjusting semi-fixed variable resistors

5. Test Disc SZZP1014F or SZZP1054C and Black Band Disc SZZP1057
6. QRB Power cord ass'y YESFZ5017

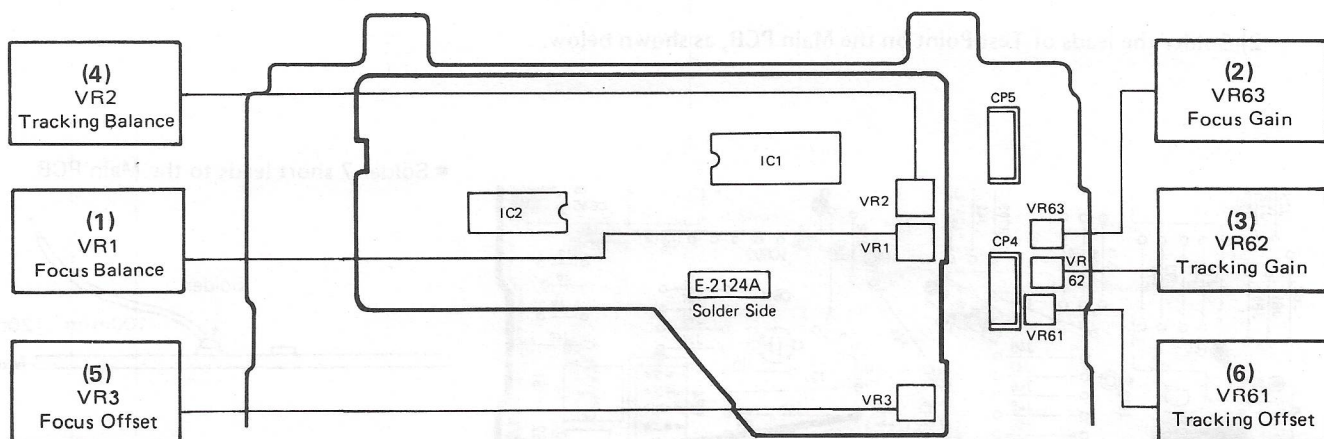
Caution: Use of controls making adjustments, or performing procedures other than specified herein may result in hazardous exposure to radiation.

A5-1-2 Alignment Procedures and Points

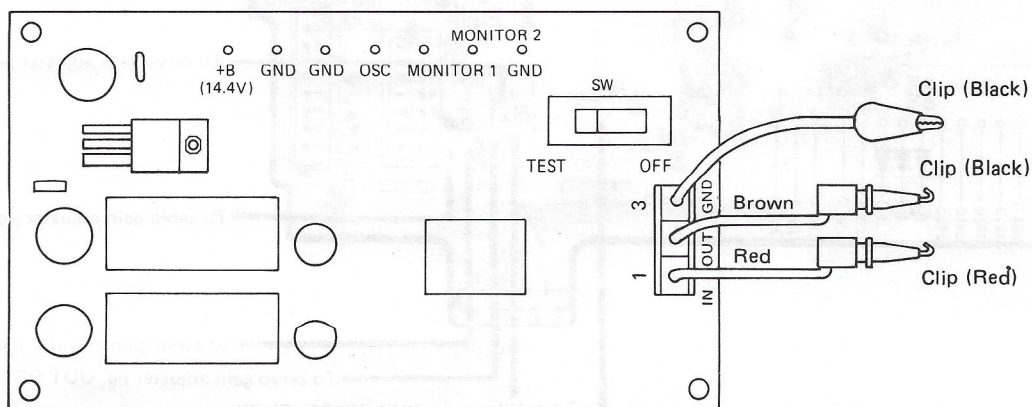
Remove the upper cover and bottom cover, solder the leads to test points, and follow the sequence below.

Note: Do not attempt alignments when the deck is disconnected.

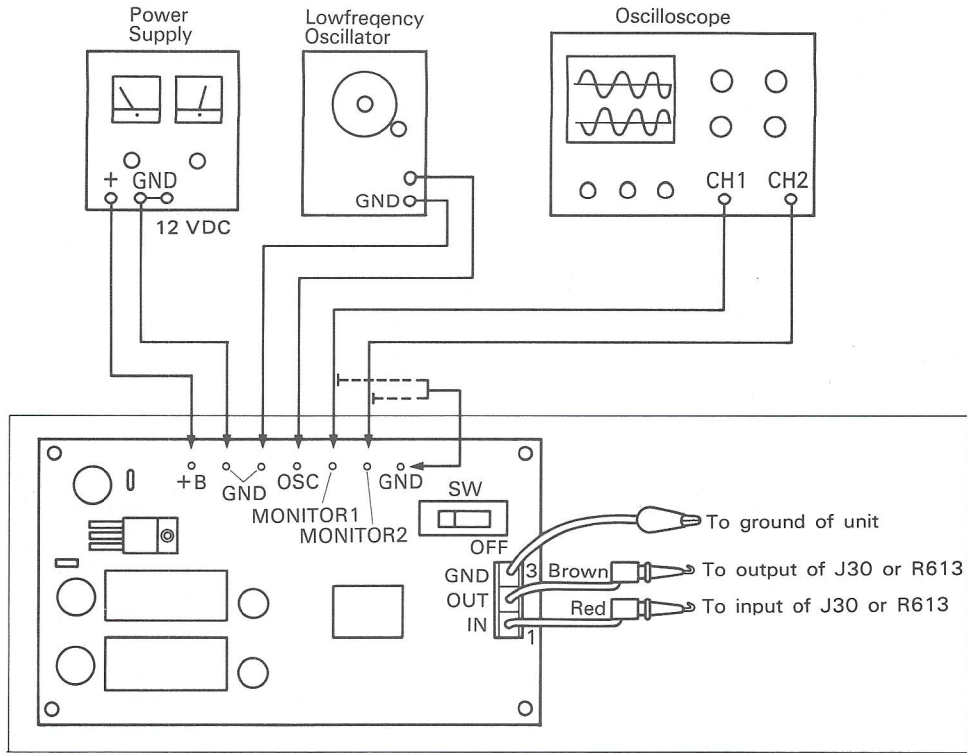
- | | | |
|-------------------------------|------|--|
| 1) Focus balance alignment | VR1 | [TP (IC601-Pin12) (ARF)] |
| 2) Focus gain alignment | VR63 | [TP (IC601-Pin9) (FE OUT)
TP (IC1-Pin16) (FPI)] |
| 3) Tracking gain alignment | VR62 | [TP (IC601-Pin6) (TE OUT)
TP (IC1-Pin17) (TPI)] |
| 4) Tracking balance alignment | VR2 | [TP (IC601-Pin6) (TE OUT)
TP (IC1-Pin17) (TP1)] |
| 5) Focus offset alignment | VR3 | [TP (IC601-Pin12) (ARF)
TP (IC601-Pin8) (FEG)] |
| 6) Tracking offset alignment | VR61 | [TP (IC601-Pin12) (ARF)
TP (IC601-Pin5) (TEG)] |



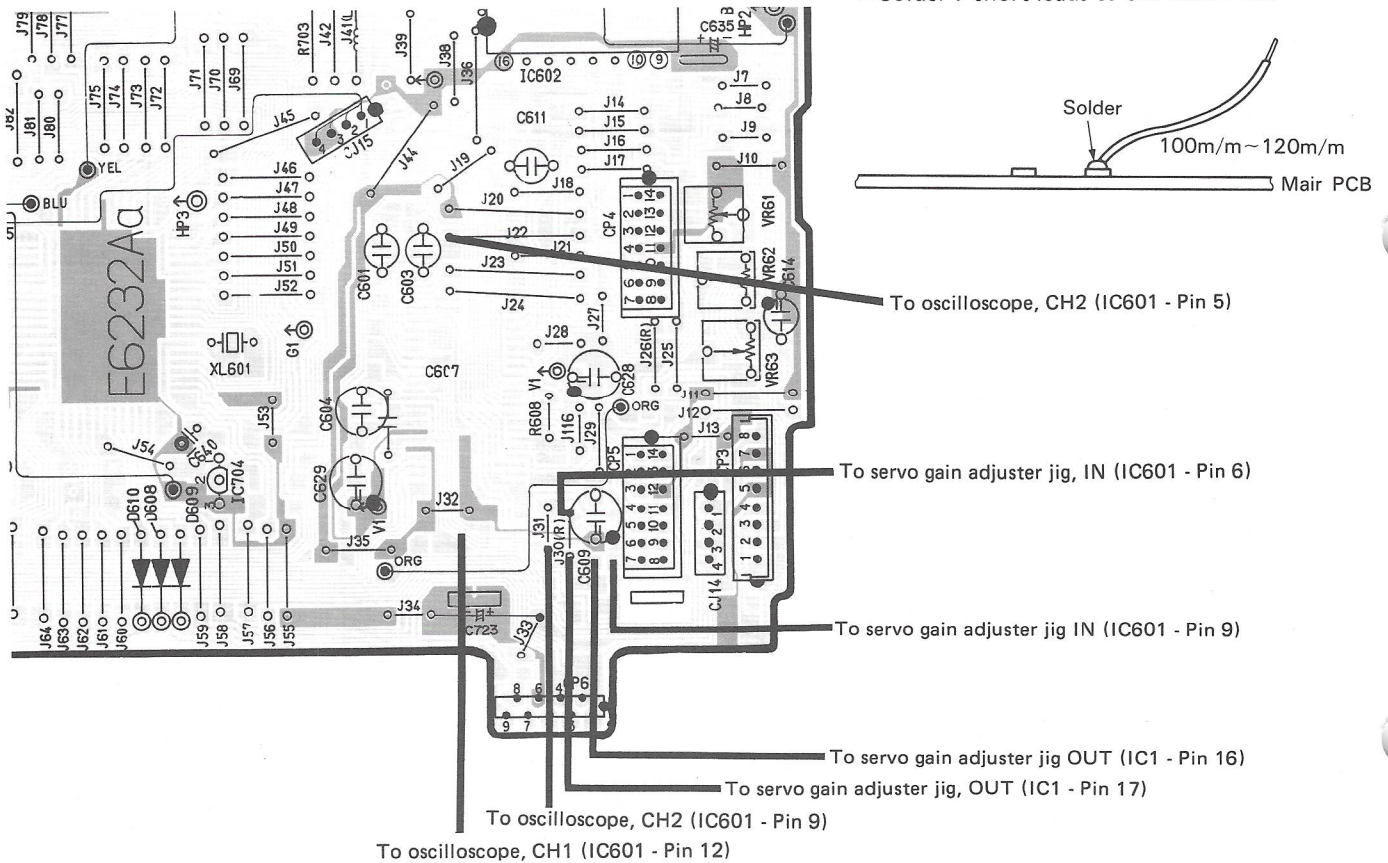
A5-1-3 CD Gain Alignment Jig



1) Wiring of Servo Gain Adjuster



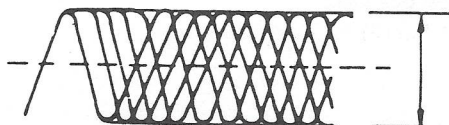
2) Solder the leads of Test Point on the Main PCB, as shown below.



A5-1-4 Adjusting Methods

1) Focus balance alignment (Best eye)

- Test point : IC601-Pin12.
- Oscilloscope setting : Volt 20 mV. Sweep 0.5 μ S Input selector AC
- Operation : Play test disc track # 1.
- Alignment : RF signal appears. Adjust VR1 until the eye pattern extends widest.
- Waveform sample :

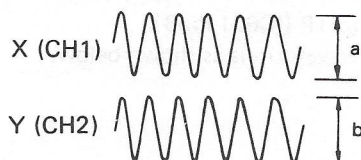


Adjust to fully widen the waveform

2) Focus gain alignment (Use of CD gain alignment jig)

- Test point : Both ends of J30
-
- CD gain alignment jig connection
 - 1) Clip Chassis
 - 2) Lead IN (Red) J30 IN (IC601-Pin9)
 - 3) Lead OUT (Brown) J30 OUT (IC1 Pin-16)
 - 4) OSC terminal Low frequency OSC
 - 5) Monitor Ch1, Ch2 terminals Oscilloscope probe Ch1, Ch2
 - Low frequency oscillator setting
 - 1) Frequency 1 kHz
 - 2) Level 200 mVp-p
 - Oscilloscope setting (Ch1, Ch2) :

Volt 20 mV Sweep 0.5 ms Input selector AC
 - Operation : Set the lever switch on the CD gain alignment jig to TEST. Insert a disc and play.
 - Alignment : Adjust VR63 until both channels are equal in amplitude.
 - Waveform sample :

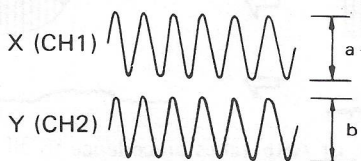


(Adjust to a = b)

3) Tracking gain alignment (Use of CD gain alignment jig)

- Test point : Both ends of R613
-
- CD gain alignment jig connection
 - 1) Clip Chassis
 - 2) Lead IN (Red) R613 IN (IC601-Pin6)
 - 3) Lead OUT (Brown) R613 OUT (IC1-Pin17)
 - 4) OSC terminal Low frequency OSC
 - 5) Monitor 1, Monitor 2 terminals Oscilloscope probe Ch1, Ch2
 - Low frequency oscillator setting
 - 1) Frequency 1 kHz
 - 2) Level 400 mVp-p
 - Oscilloscope setting (Ch1, Ch2) :

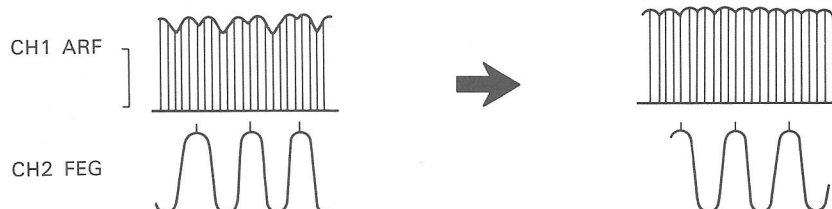
Volt 20 mV Sweep 0.5 ms Input selector AC
 - Operation : Set the lever switch on the CD gain alignment jig to TEST. Insert a disc and play.
 - Alignment : Adjust VR62 until both channels are equal in amplitude.



(Adjust to a = b)

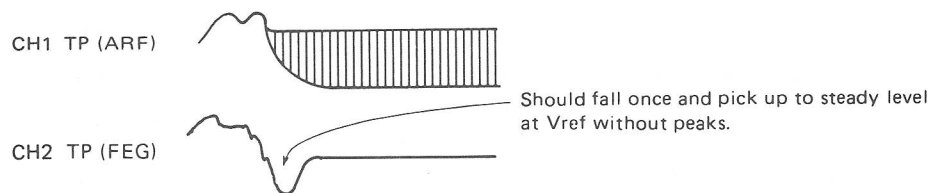
4) Tracking balance alignment (Use of CD gain alignment jig)

- Test point : Both ends of R613
- CD Adjusting jig connection :
 1. Clip Chassis
 2. Lead IN (Red) R613 IN (IC601-Pin6)
 3. Lead OUT (Brown) R613 OUT (IC1-Pin17)
 4. Monitor 2 Oscilloscope CH2
 5. Connect CH1 of oscilloscope to TP (IC601-Pin12)
- Low Frequency Oscillator setting
 - Frequency 1 kHz
 - Level 400mVp-p
- Oscilloscope setting : Volt 20 mV/div. Sweep 1ms/div. (Normal) Input Selector AC
- Operation : Set the level switch on the CD gain alignment jig to TEST. Insert a test disc and play.
- Alignment : Adjust VR2 until the peaks of CH1 (ARF) envelope evenly cover the peaks and bottoms of CH2 (TEG).



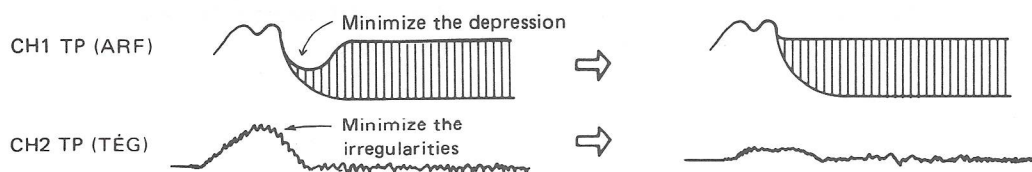
5) Focus offset alignment

- Test point : TP (ARF) (IC601-Pin12), TP (FEG) (IC601-Pin8)
- Oscilloscope setting
 - CH1 : Volt 20 mV/div. Sweep 0.5 ms/div. (Normal)
 - CH2 : Volt 5 mV/div. Sweep 0.5 ms/div. (Normal)
- Operation : Play the seventh track of disc (CDT - 002). Connect CH1 of oscilloscope to TP (IC601-Pin12) and CH2 of the same to TP (IC601-Pin8).
- Alignment : Adjust VR3 until the waveform is as shown below.
- Waveform Sample :



6) Tracking offset alignment

- Test point : TP (ARF) (IC601-Pin12), TP (TEG) (IC601-Pin5)
- Oscilloscope setting (CH1, CH2) :
 - Volt 20 mV/div. Sweep 0.2 ms/div. (Normal) Input Selector AC
- Operation : Play the seventh track of black band disc. Connect CH1 of oscilloscope to TP (IC601-Pin12) and CH2 to TP (IC601-Pin5).
- Alignment : Adjust VR61 to minimize the depression (CH 1) and to make the FE signal as flat as possible (CH 2).



Note: Eliminating the groove of ARF takes precedence to all others.